Project Phase 1

ITCS 371

Introduction to Software Engineering

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Semester 1/2024

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Introduction

Travel Naja is a new travel booking platform that aims to make trip planning easy and complete for users. Our client wants to stand out in the busy travel booking market by offering a strong system that can compete with well-known platforms like Booking.com and Agoda. Travel Naja offers three main services: flight bookings, car rentals, and hotel reservations. It also has a special feature that lets users book local tour guides, adding a personal touch to their trips.

Unlike old-style travel booking systems that manage their own flights, cars, and hotel rooms, Travel Naja works with existing booking agencies. This approach makes things easier for the client and gives users more choices from trusted providers. By using APIs to get and show real-time information from these agencies, Travel Naja makes sure users have the latest details to make good choices based on what they like.

The local guide booking feature is what makes Travel Naja different from other platforms. The client manages this feature themselves, keeping a list of good local guides. To make sure users have a great experience, all local guides must meet certain standards before they can be on the platform. People who want to be guides must fill out a form about their skills, knowledge, and tour styles. The back-end staff then checks and approves these forms. This careful process ensures that users get knowledgeable and professional guides, making their trips more real and personal.

Travel Naja is built to handle lots of users and bookings at the same time. This is important because many people use travel booking services, especially during busy travel seasons. The system can deal with thousands of bookings every second without slowing down. To do this, it uses advanced ways to balance the workload and makes its database searches faster, so users have a smooth experience.

Keeping users' information safe is very important for Travel Naja. With more online threats these days, it's crucial to protect users' personal and payment details. The platform uses the latest security technology to keep data safe when it's sent between the platform and other

agencies. It also uses secure payment systems so users can book their flights, cars, and hotels without worrying.

Travel Naja knows that different travelers have different needs. That's why the system works well on both computers and mobile devices. Whether users prefer to book trips on their desktop or on the go, the system is easy to use. There are special apps for iPhones and Android phones, making it convenient for mobile users. The desktop version is also user-friendly, with a design that looks good on any screen size.

In summary, Travel Naja is a complete travel booking platform that combines the best parts of existing services with new features for modern travelers. By working with established booking agencies, offering a high-quality local guide service, and focusing on security, growth, and user experience, the platform is ready to make a big impact in the travel booking market. Whether on a computer or phone, Travel Naja gives users a reliable and enjoyable way to plan and book their trips, setting a new standard for travel booking platforms.

Functional Requirements

R1: The system shall allow users to register as members, enabling access to personalized booking features and saved preferences.

R2: The system shall allow members to log in with their own accounts.

R3: The system shall provide flight booking functionality, allowing users to search for, select, and book flights from various airlines by using APIs.

R4: The system shall offer car rental booking functionality, allowing users to select vehicle types, rental times, and pickup/drop-off locations from multiple rental agencies.

R5: The system shall provide hotel booking functionality, allowing users to search for and book different types of rooms from various hotels.

R6: The system shall allow local guides to apply to join the platform by submitting an application form that details their qualifications, expert area, and tour offerings.

R7: Backend staff shall be able to review, approve, or reject local guide applications based on the information provided in the application form.

R8: Guide bookings shall have a capacity limit and be associated with the order system to prevent overbooking.

R9: The system shall integrate with flight booking agencies' APIs to retrieve real-time flight data, manage availability, and process reservations.

R10: The system shall integrate with car rental agencies' APIs to retrieve real-time vehicle availability, manage inventory, and process reservations.

R11: The system shall integrate with hotel booking agencies' APIs to retrieve room data and process reservations.

R12: The system shall provide credit card payment functionality and securely communicate with the bank's credit card system.

R13: The system shall offer user profile management features, including payment information and travel preferences.

R14: The system shall offer advanced search functionality, allowing users to perform comprehensive searches based on destination and schedule.

R15: The system shall provide travel recommendation features, suggesting flights, cars, and hotels based on user preferences.

R16: The system shall support a multi-language interface to accommodate global users.

R17: The system shall offer reporting management features, providing backend staff with various statistics and reports.

R18: The system shall allow users to view their booking history.

R19: The system shall allow users to manage their own booking history.

R20: The system shall allow staff to publish promotions about flight ticket booking, car rental, hotel booking, and local guide booking.

Non-functional Requirements

NR1: The system should have a user-friendly interface to help users across all platforms, including desktop and mobile applications.

NR2: The system shall support thousands of concurrent requests per second to meet the access needs of tens of thousands of users per month.

NR3: Communication between the system, third-party APIs, and payment platforms shall use advanced encryption technologies to ensure data security.

NR4: The desktop website shall ensure compatibility with mainstream browsers and have a responsive design.

NR5: The system shall be optimized for performance, ensuring fast loading times and minimal response delay.

NR6: The system shall be highly scalable, allowing for easy expansion to accommodate increased user load and additional functionalities as needed.

NR7: The system shall ensure high availability, with minimal downtime, and include failover mechanisms to maintain service continuity during outages.

NR8: The system shall maintain the integrity and consistency of data across all modules, ensuring accurate and reliable information is presented to users.

NR9: The system shall include robust logging and monitoring features, enabling backend staff to track system performance, identify issues, and ensure compliance with security protocols.

NR10: The system shall adhere to industry-standard accessibility guidelines, ensuring that it is usable by individuals with disabilities and complies with relevant legal requirements.

NR11: The system shall support multiple platforms, including a desktop website and native mobile applications for iOS and Android.

Actors

A1: Member users: Can access the platform's various booking functions, including flight, car rental, hotel, and local guide bookings. They can also manage their personal information, payment methods, and booking history, including the ability to modify or cancel reservations.

A2: Local guides: Can apply to offer local guide services through the platform. They can submit applications to be approved by the backend staff, and once approved, they can manage their tour plans, including updating tour details and availability.

A3: Backend group: They are responsible for managing the platform’s operations. Their duties include reviewing and approving local guide applications, generating and analyzing various statistical reports, managing promotions, and ensuring the overall functionality and security of the system.

A4: Frontend group: Focused on creating and maintaining the user interface and user experience aspects of the platform. They are responsible for designing and implementing new features, ensuring that the interface is intuitive, responsive, and accessible across all supported devices and platforms.

A5: Agencies' APIs : Providing real-time access to flight, car rental, and hotel booking information from trusted third-party providers. These APIs allow Travel Naja to display up-to-date availability, pricing, and options without managing its own inventory.

A6: Bank's Credit Card System: The Bank's Credit Card System is responsible for securely processing payments made by users during flight, car rental, hotel, and guide bookings. It handles the verification of user credit card details and ensures the successful completion of transactions.

Use Case Diagram

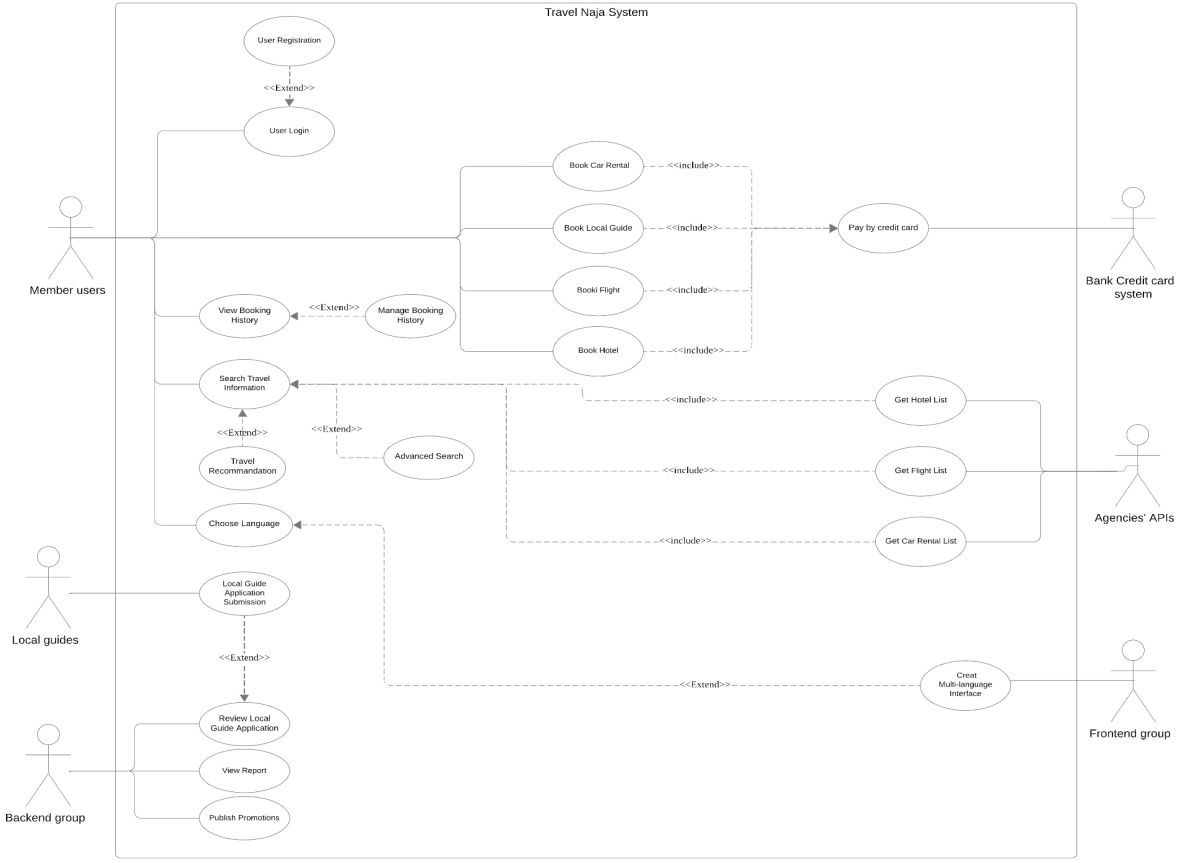

Figure 1.1 Image of Use Case Diagram

https://lucid.app/lucidchart/c04b0765-43fa-49c8-bd89-0929b8247f34/edit?viewport_loc=1148%2C-234%2C4080%2C2358%2C0_0&invitationId=inv_b478872d-7862-4aaa-a68d-b2e7aa5517ab

Use Case Narrative

Use Case Name	Book Car Rental
Goal in Context	To allow a user to rent a car through the Travel Naja platform by accessing available car rental options via external booking agencies.
Primary Actor	User(member user)
Secondary Actor	● car rental Agencies' APIs ● Bank/Credit card payment system
Precondition	● The user has an account and is logged into the system. ● Car rental agencies must have available cars. ● The payment system must be read to process the transaction.
Trigger	The user uses the "book car rental" .
Scenario (Typical Flows of Events)	1. The user logs into the travel Naja platform. 2. The user selects the "book car rental". 3. The system retrieves the list of available cars from the connected car rental agencies via APIs. 4. The user reviews the options and selects a preferred car. 5. The system redirects the user to a payment interface. 6. The user completes the payment by using a credit card. 7. The system confirms the car rental booking and sends a confirmation message to the user.

Exceptions	● The car rental booking is successfully confirmed,and the user receives a confirmation message. ● Switch to other payment methods after payment failure and automatically detect.
Post-condition	The booking details are stored in the user's booking history.

Use Case Name	Search travel information
Goal in Context	To allow users to search for travel-related information,including flights, hotels,car rentals,and local guides,through the Travel Naja platform.
Primary Actor	User (member user)
Secondary Actor	● Agencies' APIs (for flights, hotels, car rentals) ● Local guide list managed by the backend group
Precondition	● The user must be logged into the system. ● The system has access to external travel information through APIs.
Trigger	The user selects the “Search travel information”, they want to know the information about travel.
Scenario (Typical Flows of Events)	1. The user logs into the Travel Naja platform. 2. The user selects "Search Travel Information." 3. The system displays available categories for flights, car rentals, hotels, and local guides. 4. The user selects the type of information they want to search (e.g., flights, hotels, cars, local guides). 5. The system retrieves and displays the relevant options from external agencies or the internal guide list. 6. The user reviews the information and selects an option to proceed with a booking or further inquiry.

Exceptions	● The relevant travel information is displayed to the user. ● If there are no available results matching the user's criteria, the system suggests alternate options or informs the user that no results were found.
Post-condition	The user can choose to proceed with booking or refine their search.

Use Case Name	Local guide application submission
Goal in Context	To allow local guides to submit an application to be listed on the Travel Naja platform, providing information about their skills, knowledge, and tour styles.
Primary Actor	Local Guide
Secondary Actor	Backend Group
Precondition	● The local guide has registered or created an account on the Travel Naja platform. ● The platform is already accept in application.
Trigger	Assist local tour guides in logging information and data into the platform for easy booking.
Scenario (Typical Flows of Events)	1. The local guide logs into the Travel Naja platform. 2. The local guide navigates to the "Local Guide Application Submission" section. 3. The system presents the guide with a form to fill out, requiring details like skills, knowledge, tour styles, and availability. 4. The guide fills out the form and submits it. 5. The system stores the application and notifies the backend group.

Exceptions	● The local guide's application is successfully submitted and stored in the system. ● Successfully recorded the information of local tour guides into the database and achieved results.
Post-condition	The backend group receives the application for review.

Context Diagram (DFD Level 0)

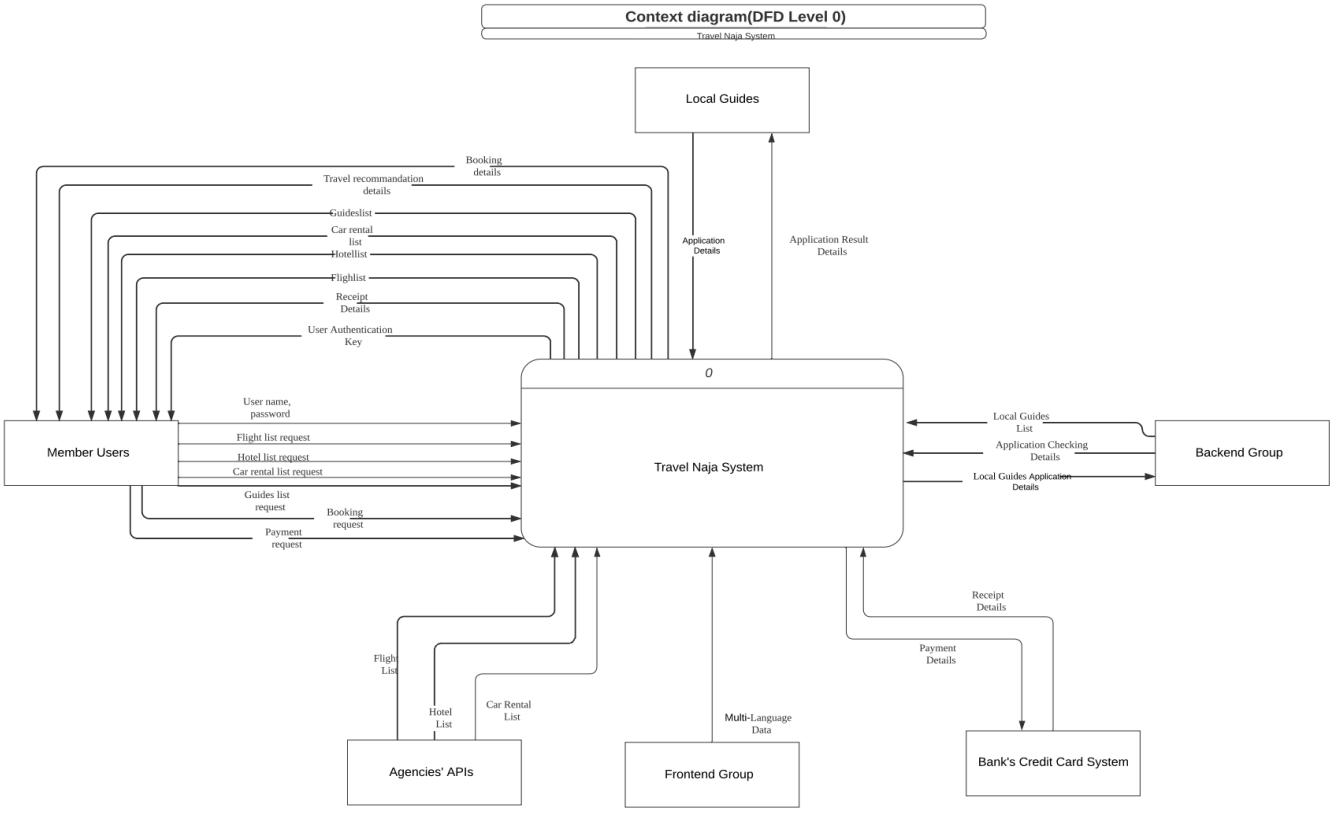

Figure 1.2 Image of Context Diagram (DFD Level 0)

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DFD Level 1

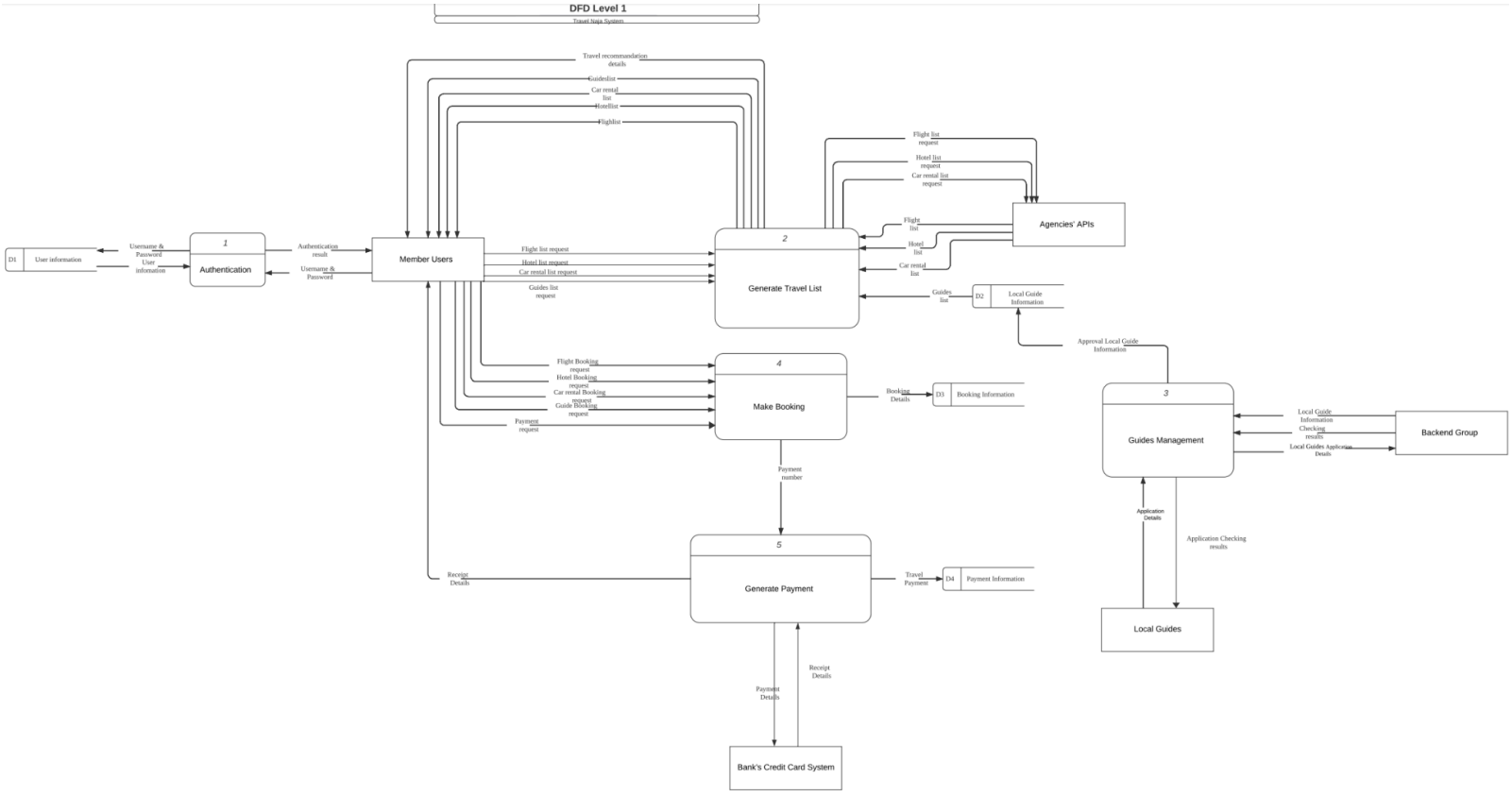

Figure 1.3 Image of Context Diagram (DFD Level 1)

https://lucid.app/lucidchart/ac39addf-97cc-43e9-be4f-2ed54150e66d/edit?viewport_loc=978%2C-166%2C1843%2C943%2C0_0&invitationId=inv_8d0f4a9d-5763-44ed-bbc8-6308c73b2c3d